$$= 2\sqrt{x^2+x+1}+2\log\left|x+\frac{1}{2}+\sqrt{\left(x+\frac{1}{2}\right)^2+\left(\frac{\sqrt{3}}{2}\right)^2}\right|+c$$

Therefore, $I = 2\sqrt{x^2+x+1}+2\log\left|x+\frac{1}{2}+\sqrt{x^2+x+1}\right|+c$

(iv) $\int \frac{5x-7}{\sqrt{3x-x^2-2}}dx$

$$5x-7 = A\frac{d}{dx}(3x-x^2-2)+B$$

$$5x-7 = A(3-2x)+B$$

Comparing the coefficients of like terms, we get

$$-2A = 5 \Rightarrow A=-\frac{5}{2}; 3A+B=-7 \Rightarrow B=\frac{1}{2}$$

$$I = \int \frac{\frac{-5}{2}(3-2x)+\frac{1}{2}}{\sqrt{3x-x^2+2}}dx$$

$$I = -\frac{5}{2}\int \frac{3-2x}{\sqrt{3x-x^2+2}}dx+\frac{1}{2}\int \frac{1}{\sqrt{3x-x^2+2}}dx$$

$$= \left(-\frac{5}{2}\right)2\sqrt{3x-x^2+2}+\frac{1}{2}\int \frac{1}{\sqrt{\left(\frac{\sqrt{17}}{2}\right)^2-\left(x-\frac{3}{2}\right)^2}}dx$$

$$= -5\sqrt{3x-x^2+2}+\frac{1}{2}\sin^{-1}\left(\frac{x-\frac{3}{2}}{\frac{\sqrt{17}}{2}}\right)+c$$

Thus, $I = -5\sqrt{3x-x^2+2}+\frac{1}{2}\sin^{-1}\left(\frac{2x-3}{\sqrt{17}}\right)+c$

EXERCISE 11.11

Integrate the following with respect to x :

(1) (i) $\frac{2x-3}{x^2+4x-12}$ (ii) $\frac{5x-2}{2+2x+x^2}$ (iii) $\frac{3x+1}{2x^2-2x+3}$

(2) (i) $\frac{2x+1}{\sqrt{9+4x-x^2}}$ (ii) $\frac{x+2}{\sqrt{x^2-1}}$ (iii) $\frac{2x+3}{\sqrt{x^2+4x+1}}$

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